Huawei Access Network SOP for KOR-ACC-83 Routing Protocol Flap Failure

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2886

Description

This Standard Operating Procedure (SOP) addresses the critical failure scenario of a Huawei Access Network node (KOR-ACC-83) experiencing a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) at the Koramangala Datacenter. The failure is attributed to the complex interplay of the Huawei's proprietary routing engine, the egress interface's hardware capabilities, and the OSPF process's reconfiguration frequency. Specifically, the Huawei Access Network node's egress interface, which is configured as a transit interface for both OSPF and IS-IS routing protocols, experiences increased CPU utilization and packet forwarding latency due to the high BGP route churn and OSPF adjacency loss. This exacerbates the likelihood of ISIS LSP flooding, further degrading network stability and reliability. In this scenario, the Huawei Access Network node's CPU utilization exceeds 85% for an extended duration, resulting in a Critical alarm (ALARM_2886). To mitigate this failure scenario, it is essential to promptly identify and resolve the root cause of the routing protocol flap, ensuring the timely restoration of network stability and minimizing the risk of service disruptions or data loss.

Pre-requisite Checks

- Verify the Koramangala Datacenter's environmental conditions, including temperature, humidity, and power supply stability.

- Confirm the redundant link status and ensure that it is functioning correctly.

- Dispatch remote hands to the Koramangala Datacenter for on-site support, if necessary.

Potential Root Causes

Cause	Probability
High CPU utilization due to excessive BGP route churn and OSPF adjacency loss.	High
Inadequate egress interface capabilities, leading to packet forwarding latency.	Medium
ISIS LSP flooding due to OSPF adjacency loss.	Low

Diagnostic Commands & Expected Outputs

```
Command: display cpu statistics | include ' utilization ', Expected Output: CPU utilization: 90.00%, Description: This command isolates the CPU utilization of the Huawei Access Network node, providing insight into the system's CPU load and potential bottlenecks.

Command: display bgp routing-table | include 'route churn ', Expected Output: Route churn: 1000 routes changed in 5 minutes, Description:
```

```
This command identifies the BGP route churn and its frequency, helping to diagnose the root cause of the CPU utilization issue.

Command: display ospf neighbor | include ' adjacency loss ', Expected Output: Adjacency loss: 3 neighbors lost in 10 minutes, Description: This command examines the OSPF adjacency loss and its frequency, providing further insight into the potential root cause of the CPU utilization issue.
```

Resolution Steps

Step 1: Step: Optimize OSPF process configuration to reduce adjacency loss., Command: ospf process-id optimize adjacency-loss, Description: This command optimizes the OSPF process configuration to minimize adjacency loss, thereby reducing CPU utilization and packet forwarding latency.

Step 2: Step: Upgrade egress interface hardware to improve packet forwarding capabilities., Command: upgrade interface egress interface-name, Description: This command upgrades the egress interface hardware to improve packet forwarding capabilities, reducing latency and CPU utilization.

Step 3: Step: Adjust BGP route churn by redistributing routes., Command: bgp redistribute route-churn, Description: This command adjusts the BGP route churn by redistributing routes, reducing the CPU utilization and packet forwarding latency.

Post-Check Verification

```
Command: display cpu statistics | include ' utilization ', Expected Output: CPU utilization: 20.00%, Description: This command verifies the CPU utilization of the Huawei Access Network node, ensuring that it has returned to a stable and acceptable level.

Command: display bgp routing-table | include 'route churn ', Expected Output: Route churn: 100 routes changed in 5 minutes, Description: This command verifies the BGP route churn, confirming that it has been reduced to an acceptable level.

Command: display ospf neighbor | include ' adjacency loss ', Expected Output: Adjacency loss: 0 neighbors lost in 10 minutes, Description: This command verifies the OSPF adjacency loss, confirming that it has been eliminated.
```

Roll-back Procedures

- Step: Revert OSPF process configuration to its original state., Command: ospf process-id revert, Description: This command reverts the OSPF process configuration to its original state, restoring the original routing behavior.

- Step: Downgrade egress interface hardware to its original version., Command: downgrade interface egress interface-name, Description: This command downgrades the egress interface hardware to its original version, reverting to the original packet forwarding capabilities.

- Step: Revert BGP route churn by redistributing routes., Command: bgp redistribute route-churn revert, Description: This command reverts the BGP route churn by redistributing routes, restoring the original routing behavior.